

Week 8 Task Report

Student Performance Dashboard

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Introduction

Week 8 task focused on developing an interactive Student Performance Dashboard using Power BI. The dashboard was created to analyse students' academic performance through key performance indicators, subject-wise comparisons and different student-related factors. Interactive slicers and a drill-through Academic Profile page were also implemented to make the analysis more dynamic and user-friendly.

Objective

The objective of Week 8 was to create an interactive Student Performance Dashboard using Power BI. The dashboard was designed to analyse students' academic performance based on subject scores, parental education, test preparation, lunch type and grades. Interactive slicers and a drill-through page were also created to make the dashboard easier to explore.

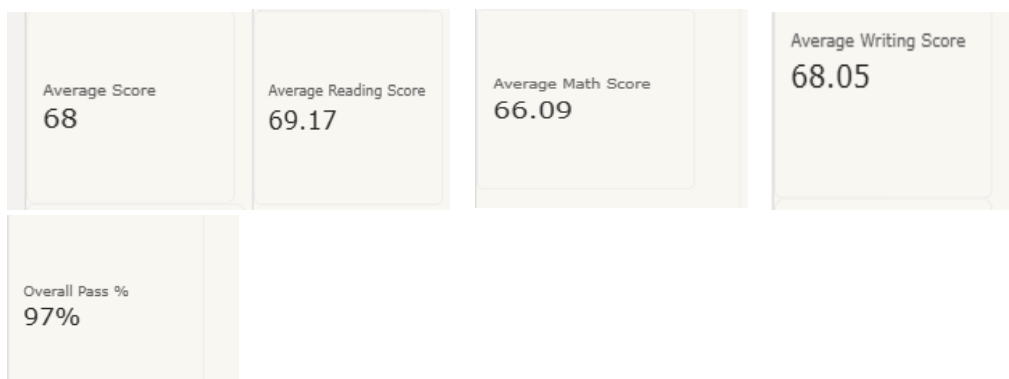
KPI Cards

Five KPI cards were created to provide a quick overview of student performance:

- **Average Score:** 68
- **Average Math Score:** 66.09
- **Average Reading Score:** 69.17
- **Average Writing Score:** 68.05
- **Overall Pass Percentage:** 97%

These cards provide a summary of the overall academic performance and individual subject performance.

KPI Cards Showing Overall Student Performance

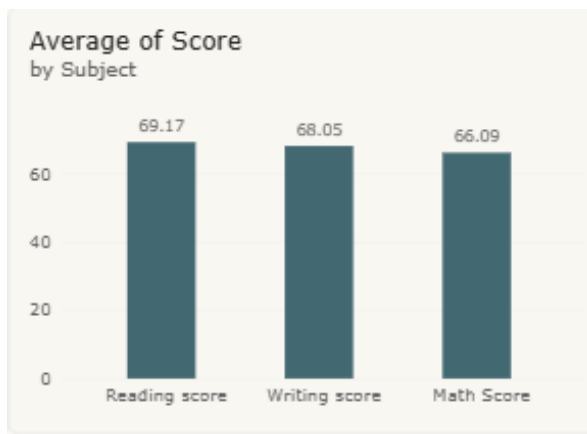


Subject Performance Analysis

A **clustered column chart** was created to compare the average scores of students in the three subjects: **Math, Reading and Writing**. The Student Marks data was used to analyse the subject-wise scores, and the scores were summarized using the **Average** aggregation.

The analysis shows that **Reading has the highest average score of 69.17**, followed by **Writing with 68.05**, while **Math has the lowest average score of 66.09**. This visual makes it easier to compare the academic performance across the three subjects.

Average Score by Subject

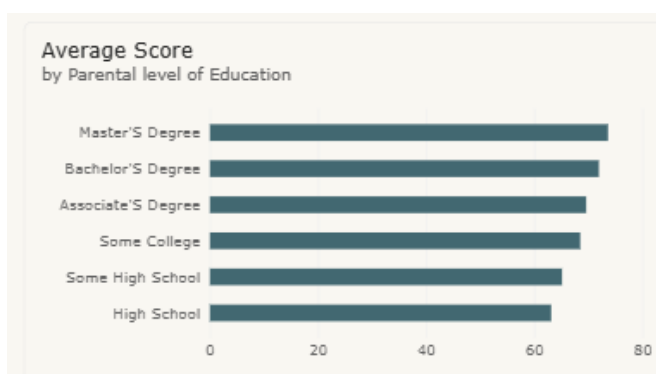


Parental Education Analysis

A **clustered bar chart** was created to study the relationship between **parental education level and students' average scores**. The parental education categories include Master's Degree, Bachelor's Degree, Associate's Degree, Some College, Some High School and High School.

The chart allows the average performance of students from different parental education groups to be compared easily. This provides a demographic perspective of the student performance data and helps identify differences between the groups.

Average Score by Parental Education Level

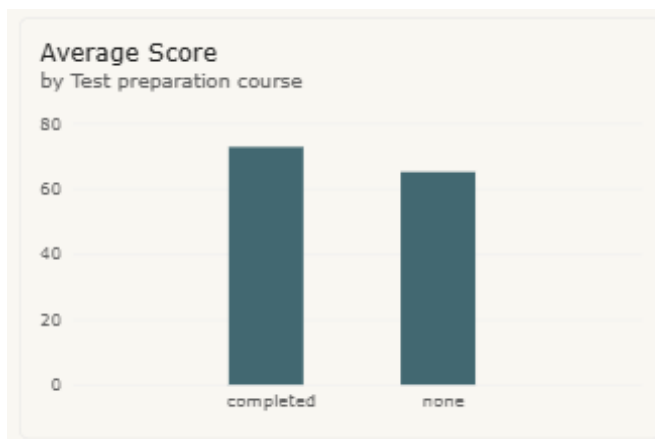


Test Preparation Analysis

A clustered column chart was created to compare the academic performance of students based on their test preparation course status. The two categories considered were Completed and None.

The average score was used to compare the performance of the two groups. A higher average score can be observed for the group that completed the preparation course. The visual helps identify differences in student performance based on test preparation status without assuming that the preparation course directly caused the difference.

Average Score by Test Preparation Course

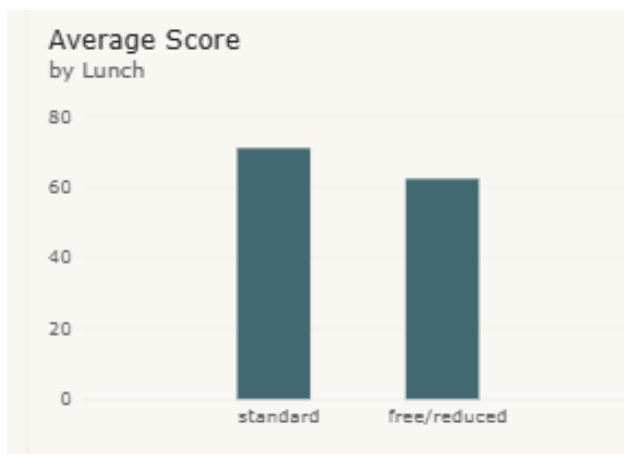


Lunch Type Analysis

A visual was created to analyse student performance based on Lunch Type. The dashboard compares the average scores of students receiving different lunch categories.

This analysis provides a comparison of academic performance across the lunch groups and helps identify patterns in the dataset. The visual makes it easier to observe which lunch category has a higher or lower average score.

Average Score by Lunch Type

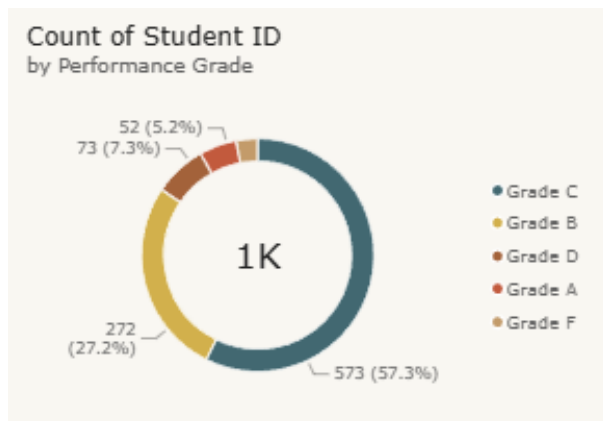


Grade Distribution

A donut chart was created to represent the distribution of students across different Performance Grade categories. The chart uses the number of students in each grade category to show how the students are distributed across the performance levels.

The visual provides a simple overview of the overall grade distribution and makes it easier to identify the proportion of students belonging to different performance categories.

Student Performance Grade Distribution



Interactive Slicers

Multiple interactive slicers were added to make the dashboard dynamic and user-friendly. The slicers allow the user to filter the dashboard based on different student characteristics.

The slicers created were Gender, Race/Ethnicity, Parental Education, Lunch Type, Test Preparation Course and Subject. When a particular value is selected, the dashboard visuals update according to the selected filter, allowing different groups of students to be analysed interactively.

Interactive Slicers for Student Performance Analysis

Race/Ethnicity

☐ group A

☐ group B

☐ group C

☐ group D

☐ group E

Parental level of Education

☐ Associate'S Degree

☐ Bachelor'S Degree

☐ High School

☐ Master'S Degree

☐ Some College

☐ Some High School

Gender

Gender

☐ female

☐ male

Lunch

☐ free/reduced

☐ standard

Test preparation course

☐ completed

☐ none

Subject

☐ Math Score

☐ Reading score

☐ Writing score

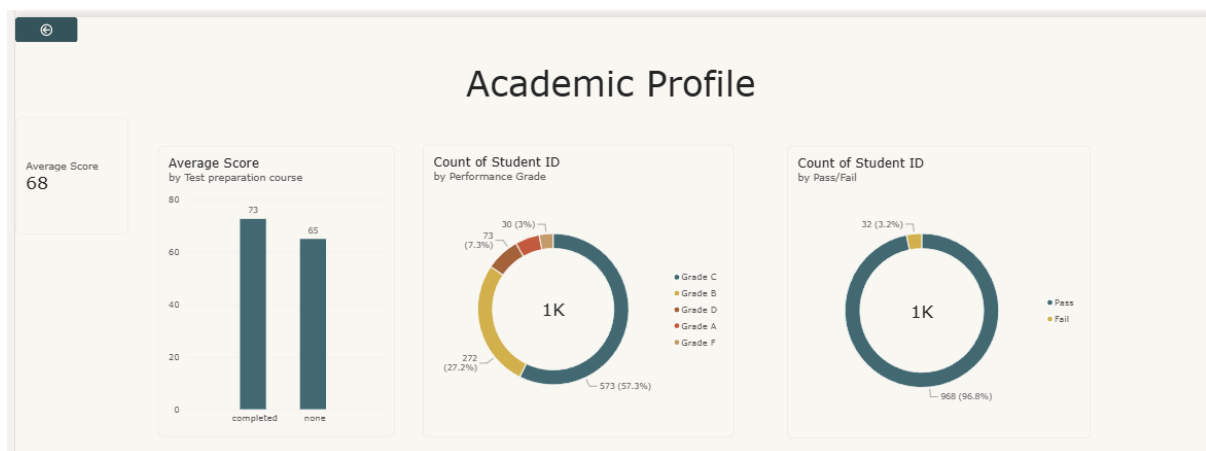
Academic Profile – Drill-through

A separate Academic Profile page was created using Power BI's drill-through functionality. The Student ID field was used as the drill-through field, allowing the user to move from the main dashboard to a detailed academic profile.

The Academic Profile page contains information related to Average Score, Grade Distribution, Pass/Fail status and Test Preparation. A Back button is also provided so that the user can easily return to the main dashboard.

This feature makes the dashboard more interactive because it allows detailed information to be viewed for a selected student rather than displaying everything on the main dashboard.

Academic Profile Drill-through Page



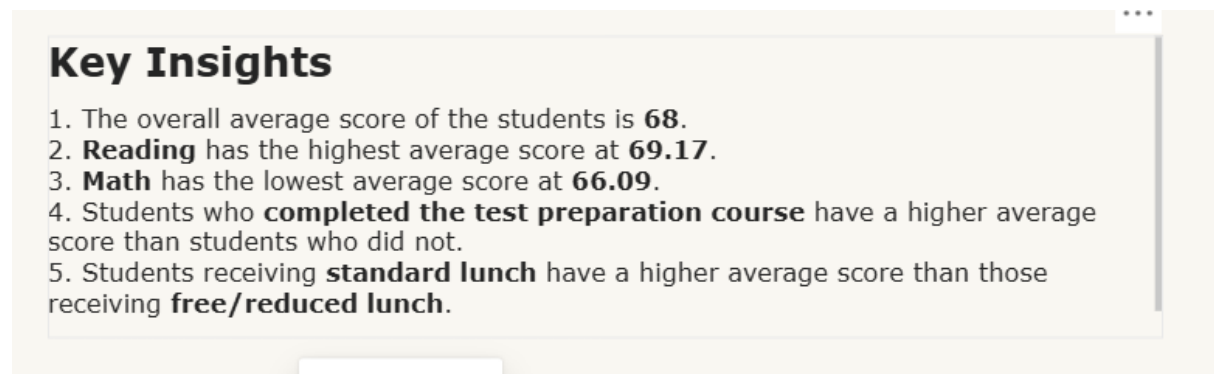
Key Insights

The dashboard provides several observations from the analysed data:

1. The overall average score of the students is 68.
2. Reading has the highest average score at 69.17.
3. Math has the lowest average score at 66.09.
4. Students who completed the test preparation course have a higher average score than students who did not.
5. Students receiving standard lunch have a higher average score than those receiving free/reduced lunch.

These observations are based on the values and patterns displayed in the dashboard. They describe the data without making unsupported causal claims.

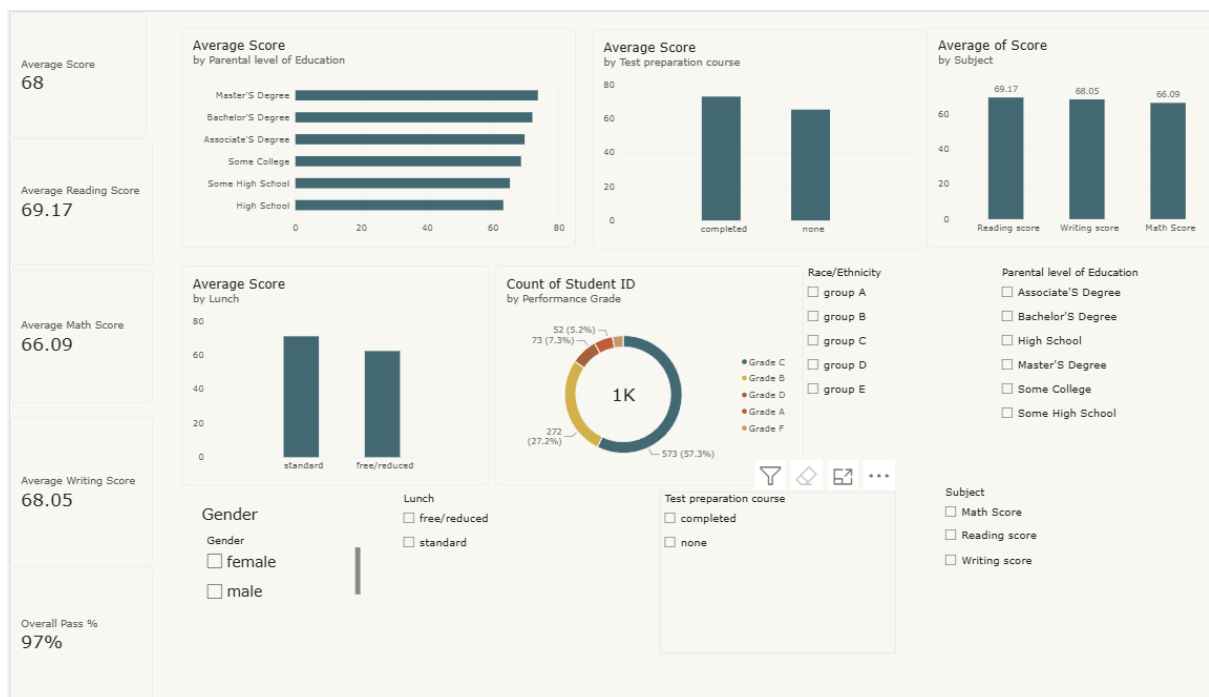
Key Insights from the Student Performance Dashboard



Main Dashboard

The main dashboard provides an overall view of student academic performance through KPI cards, subject-wise analysis, parental education analysis, test preparation analysis, lunch type analysis, grade distribution and interactive slicers.

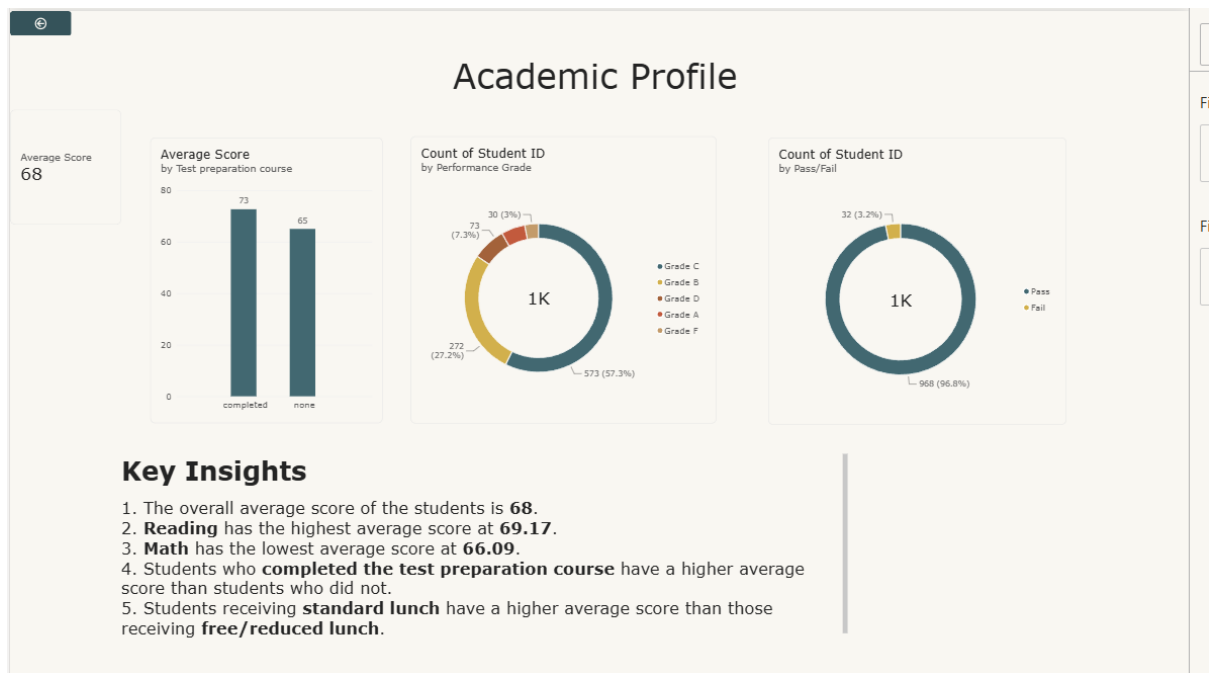
Main Student Performance Dashboard



Academic Profile

The Academic Profile page was created using the drill-through feature. It provides detailed information such as average score, grade distribution, pass/fail status and test preparation for the selected student. A Back button is available to return to the main dashboard.

Academic Profile Drill-through Dashboard



Conclusion

The Week 8 Student Performance Dashboard was successfully developed using Power BI to provide an interactive analysis of student academic performance. Different visualizations were used to analyse subject scores, parental education, test preparation, lunch type and grade distribution.

The addition of interactive slicers improved the flexibility of the dashboard, while the Academic Profile drill-through page provided a detailed view of student-level information. Overall, the dashboard presents the student performance data in a clear and interactive format and supports easier comparison and analysis of different student groups.